Name……index No……

**P525/1**

**Chemistry**

**Paper 1**

**$2^{3}/_{4}$HOURS**

**UACE EXAMINATION 2022**

S.5 CHEMISTRY

PAPER 1

**TIME: $2^{3}/_{4}$HOURS**

**INSTRUCTIONS:**

- Attempt **all** questions
- Answers must be written in the spaces provided only.
- The periodic table, with relative atomic masses is supplied at the end of the paper.
- Molar gas constant $R=8.31JK^{-1}mol^{-1}$
- Molar volume of gas at stp is 22.4 litres.

**SECTION A**

1. (a) Given the hydrides, $CH_4$,$NH_3$,$H_2O$, complete the table below. (03 marks)

| HYDRIDE | STRUCTURE | NAME |
|---|---|---|
| $CH_4$ | | |
| $NH_3$ | | |

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|---|---|---|
| $H_2O$ | | |

(b) Which hydride has the boiling point? Explain. $\left(01\frac{1}{2}\text{ marks}\right)$

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2. (a) The freezing point of a solution containing 3.294g of sulphur in 100g of naphthalene was found to be -0.830℃ and another solution containing 1.67g of iodine in the same mass of naphthalene froze at-0.84℃.

Calculate

(i) The freezing point depression constant for naphthalene(molar mass of Iodine is127).(02 marks)

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(ii) The molar mass of sulphur in naphthalene.
(02marks)

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(b

(i) Determine the molecular formula of sulphur.
$(01\frac{1}{2}\text{marks})$

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(ii) Draw the structure of sulphur . (01 mark)

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3. (a) an aqueous solution containing $2.8gdm^{-3}$ of R exerts an osmotic pressure of 380mmHg at stp

Calculate the molar mass of R

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(b)An organic compound R contains carbon, oxygen. 3.4g of R burns with a sooty flame producing $5.04dm^3$ of carbondioxide and 2.70g of water at stp

(i) calculate the empirical formula of R. ($02\frac{1}{2}$marks)

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(ii) Hence determine the molecular formula of R. (01 mark)

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(c) R reacts with a mixture of anhydrous zinc chloride and concentrated hydrochloric acid but does not react with acidified potassium dichromate(VI).

(i) Identify R (01 Mark)

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(ii) Write equation between R and a mixture of anhydrous zinc chloride and concentrated hydrochloric acid.
(01 mark)

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(d) Write equations to show how R can be converted to an alkyne.
($01\frac{1}{2}$marks)

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4. (a) the electronic configuration of element Z is: $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$

State with reasons the period and group to which the element Z belongs.
[02marks]

Period: ……

Reason:
……
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Group: ……

Reason:
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(b) (i) Write the full electronic configuration of nitrogen and phosphorus atoms.
[02marks]

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(ii) Explain why phosphorus(V)chloride can be formed but not nitrogen(V) chloride.
[01mark]

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5. The freezing point of pure benzene is 5.53℃. when 0.775g of an allotrope of phosphorus was dissolved in 50.0g of benzene the solution froze at 4.89℃. [$K_f$ for benzene = 5.12℃$mol^{-1}Kg^{-1}$]

   (a) Calculate the molecular mass of phosphorus. (03 marks)

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   (b) Deduce the molecular formula of phosphorus in benzene. [P=31] (02 marks)

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6. (a) 20$cm^3$ of a gaseous hydrocarbon Z was exploded with excess oxygen. On cooling, the reduction in volume was 30$cm^3$. If the vapour density of the hydrocarbon is 13, determine the molecular formula of Z.
   (04 marks)

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   (b) Write equations to show how Z can be synthesized from propene. Indicate reagents and conditions for the reaction.
   (05 marks)

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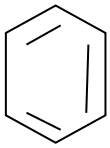
7. Complete the following equations and write the accepted mechanism for the reaction in each case.

a. 

$+ \; CH_3 - C = CH_2$ (with $CH_3$ bonded to the C) $\quad H^+ \longrightarrow$

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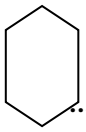
b. 

$\xrightarrow{HBr}$

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(d) Complete the following

$CH_3Br + Na \xrightarrow[\text{Heat}]{\text{Dry ether}}$

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$CH_3CH_2MgBr + H_2O \xrightarrow[\text{Heat}]{\text{Dry ether}}$

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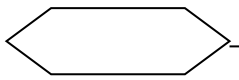

$CH_2I + Na \xrightarrow{\text{Dry ether}}$

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Heat

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Pd

$CH_2 = CH_2 + H_2 \longrightarrow$

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8. (a) State Grahams law of gaseous diffusion

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(b) 141.4cm$^3$ of gas X diffused through a porous plug in the same time as it took 50cm$^3$ of Oxygen to diffuse through the same plug under identical conditions. Calculate the relative molecular mass of X.

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9. Calculate the oxidation state of the central metal atom/ion in the following complexes

(i) $[V(CN)_6]^{3-}$

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(ii) $[CoCl_4]^{2-}$

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(i) $[Zn(OH)_4]^{2-}$

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(ii) $[Fe(CN)_4]^{2-}$

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(iii) $[Ni(CO)_4]^{2+}$

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(iv) $[NiCl_4]^{2-}$

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(v) $[CuCl_4]^{2-}$

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(vi) $[Co(NH_3)_6]^{3+}$

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(vii) $K_4Fe(CN)_6$

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(iii) $[FeSCN]^{2+}$

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10. Some half reactions and their corresponding reduction potentials are given below:

$MnO_4^-(aq) + 8H^+(aq) + 5e \longrightarrow Mn^{2+}(aq) + 4H_2O(l)$ $E^\theta = +1.5V$

$MnO_2(aq) + 4H^+(aq) + 2e \longrightarrow Mn^{2+}(aq) + 2H_2O(l)$ $E^\theta = +1.23V$

$Sn^{2+}(aq) + 2e \longrightarrow Sn(s)$ $E^\theta = +0.136V$

$Sn^{4+}(aq) + 2e \longrightarrow Sn^{2+}(aq)$ $E^\theta = +0.15V$

(a) (i) state what would be observed when excess tin powder is added to acidified solution of potassiummanganate(VII).

Explain your answer. [03 marks]

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(ii) Write equation for the reaction that takes place in (i) above.[$01\frac{1}{2}$marks]

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(b) Solid manganese (IV) oxide was added to acidified solution containing tin(II) ions.

(i) State what was observed and explain your answer.[03 marks]

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(ii) Write equation for the reaction [01$\frac{1}{2}$marks]

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11. Write equations to show how the following compounds can be synthesized.

(a) $CH_3C \equiv CCH_3$ from $CH_3CH = CH_2$ (03marks)

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(b) $(CH_3)_2CHOH$ from $CH_3CH_2Br$ $\left(02\frac{1}{2}\text{ marks}\right)$

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12. (a) Explain what is meant by the following term,

"First ionization energy"[03 marks]

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(b) State and explain how the above term in (a) varies across period 3 and down group VII.[03 marks]

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(c) Explain why:

(i) Magnesium has higher first ionization energy than aluminum. [03 marks]

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(ii) Sulphur has a lower first ionization energy than phosphorus. [03 marks]

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(i) Second electron affinity of magnesium is positive.[02 marks]

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(ii) Electron affinity of phosphorus is less negative than that of silicon.[03 marks]

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(iii) Electron affinity of fluorine is abnormally low in value. [01$\frac{1}{2}$marks]

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[Atomic numbers for Mg, Al, O, S, Si and F are 12, 13, 8, 16, 14 and 9 respectively]

13. Complete the following reactions and write the accepted mechanism.

(a) $CH_3CH_2CH = CH_2 \xrightarrow{Br_2/H_2O}$

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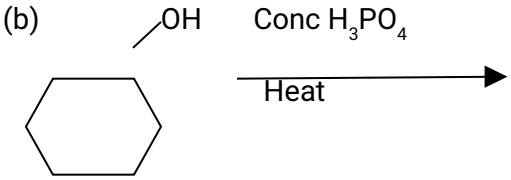
(b)

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14.

(a) What is meant by the term partition coefficient?
(02 marks)

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(b) Mention any two limitations of distribution law.
(01 mark)

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(c) Mukwano oil company is worried about the impurity X in its crude oil. One $dm^3$ of crude oil contains 5g of X. In an effort to reduce the concentration of X in oil, mukwano has discovered the secret solvent Y. The partition coefficient of X between oil and Y is 0.01. Calculate the total mass of X removed from one $dm^3$ of oil using:

(i) One portion of $100cm^3$ of solvent Y.[02 marks]

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(ii) Two $50cm^3$ portions of solvent Y.[04 marks]

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(iii) Mention any application of distribution law in the industrial chemistry? [01 mark]

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## THE PERIODIC TABLE

| 1 | 2 | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 1.0 H 1 | | | | | | | | | | | | | | | | | |
| 6.9 Li 3 | 9.0 Be 4 | | | | | | | | | | | | | | | | |
| 23.0 Na 11 | 24.3 Mg 12 | | | | | | | | | | | | | | | | |
| 39.1 K 19 | 40.1 Ca 20 | 45.0 Sc 21 | 47.9 Ti 22 | 50.9 V 23 | 52.0 Cr 24 | 54.9 Mn 25 | 55.8 Fe 26 | 58.9 Co 27 | 58.7 Ni 28 | 63.5 Cu 29 | 65.7 Zn 30 | | | | | | |
| 85.5 Rb 37 | 87.6 Sr 38 | 88.9 Y 39 | 91.2 Zr 40 | 92.9 Nb 41 | 95.9 Mo 42 | 98.9 Tc 43 | 101 Ru 44 | 103 Rh 45 | 106 Pd 46 | 108 Ag 47 | 112 Cd 48 | | | | | | |
| 133 Cs 55 | 137 Ba 56 | 139 La 57 | 178 Hf 72 | 181 Ta 73 | 184 W 74 | 186 Re 75 | 190 Os 76 | 192 Ir 77 | 195 Pt 78 | 197 Au 79 | 201 Hg 80 | | | | | | |
| 223 Fr 87 | 226 Ra 88 | 227 Ac 89 | | | | | | | | | | | | | | | |

| 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|
| 1.0 H 1 | 4.0 He 2 | | | | |
| 10.8 B 5 | 12.0 C 6 | 14.0 N 7 | 16.0 O 8 | 19.0 F 9 | 20.2 Ne 10 |
| 27.0 Al 13 | 28.1 Si 14 | 31.0 P 15 | 32.1 S 16 | 35.4 Cl 17 | 40.0 Ar 18 |
| 69.7 Ga 31 | 72.6 Ge 32 | 74.9 As 33 | 79.0 Se 34 | 79.9 Br 35 | 83.8 Kr 36 |
| 115 In 49 | 119 Sn 50 | 122 Sb 51 | 128 Te 52 | 127 I 53 | 131 Xe 54 |
| 204 Tl 81 | 207 Pb 82 | 209 Bi 83 | 209 Po 84 | 210 At 85 | 222 Rn 86 |

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|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 139 La 57 | 140 Ce 58 | 141 Pr 59 | 144 Nd 60 | 147 Pm 61 | 150 Sm 62 | 152 Eu 63 | 157 Gd 64 | 159 Tb 65 | 162 Dy 66 | 165 Ho 67 | 167 Er 68 | 169 Tm 69 | 173 Yb 70 | 175 Lu 71 |
| 227 Ac 89 | 232 Th 90 | 231 Pa 91 | 238 U 92 | 237 Np 93 | 244 Pu 94 | 243 Am 95 | 247 Cm 96 | 247 Bk 97 | 251 Cf 98 | 254 Es 99 | 257 Fm 100 | 256 Md 101 | 254 No 102 | 260 Lw 103 |

**END**

***"God hate lazy people, He loves hardworking ones!"***

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